CARBON-AWARE DATA-CENTER SCHEDULING

The Price of Sophistication: When Do Spatial and Robust Models Pay in Carbon-Aware Scheduling?

A day-ahead migration scheduler and a complexity-value decision rule

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DATA & TOOLS

Electricity Maps

Real-time grid carbon, 2021 to 2025

Solved with **CVXPY · CLARABEL · HiGHS**

Day-ahead forecast → Migration scheduler → Price each modeling layer → Decision rule

Active inter-region migration cuts worst-case emissions 4.0 to 9.9%; added sophistication (joint covariance, robustness) does not pay until a severity real grids never reach.

THE SCREEN	THE PRICE	THE LIMIT	THE LEVER
~0%	**M* = 3**	**0 / 17**	**4-10%**
value added by joint covariance and copulas	severity where robustness finally starts to pay	real grids that reach the robustness threshold	**worst-case (CVaR) emissions cut by active inter-region migration**

BACKGROUND

The promise

Modern compute is **flexible**: training and batch jobs shift in time and across data-center sites, so deferring work to **cleaner hours and regions** lowers operational carbon. Across 2021 to 2025 neighbouring grids move strongly together, so spatial structure looks like free signal a robust scheduler could turn into savings. **Question: can measured cross-region correlation actually lower out-of-sample carbon?**

DATA & SETUP

What goes in

Electricity Maps hourly carbon intensity, 2021 to 2025 (**43,824 obs/zone**); train 2021 to 2024, single **locked 2025 test**. Three US and Canadian grids spanning weak to strong residual correlation (up to 0.78). Facility: R zones × 24 h, 50 MW cap, **80% utilization**, 15 MW/h ramp, deferral window 0 to 7 h, inflexible fraction 30 to 75%.

METHOD

How it is built

A **Mahalanobis Wasserstein DRO**, a second-order cone program in **CVXPY (Clarabel/HiGHS)**, minimizing **CVaR** of carbon cost (*Rockafellar and Uryasev 2000*):

$$\min_{x\in\mathcal{X}} \ \langle \bar{\rho}, x\rangle + \varepsilon\sqrt{x^\top \hat{\Sigma} x}$$

$\langle \bar{\rho}, x\rangle$	ε	$\sqrt{(x^\top\hat{\Sigma}x)}$
expected carbon cost: the dominant mean field	ambiguity radius: how much we robustify	robustness penalty; $\hat{\Sigma}$ is the covariance — correlation enters ONLY here

Cross-region value enters **only through the off-diagonal blocks of Σ**; the pre-registered **shuffled-marginals test** zeroes exactly those blocks. We add copula and tail checks, mean-leveling tests, and walk-forward validation to 2025.

RIGOR

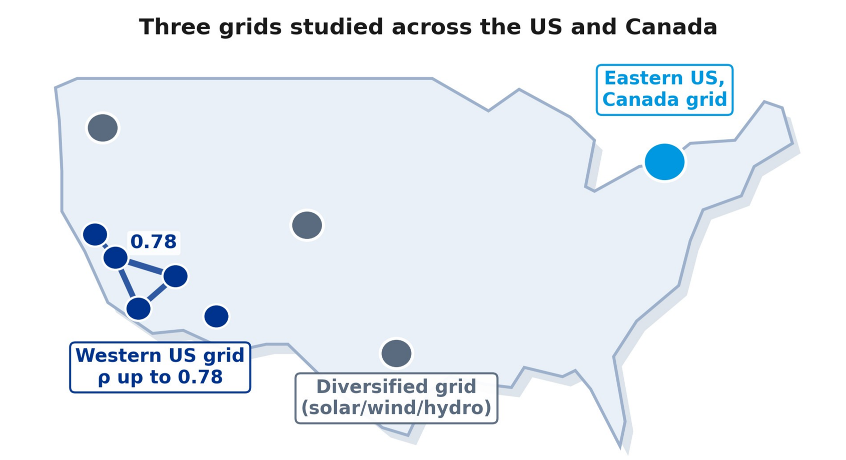
Why you can trust a null

199	unit tests, CI on every push
27	cells where TOST confirms a null effect
1	mean-dominance theorem bounding the gap

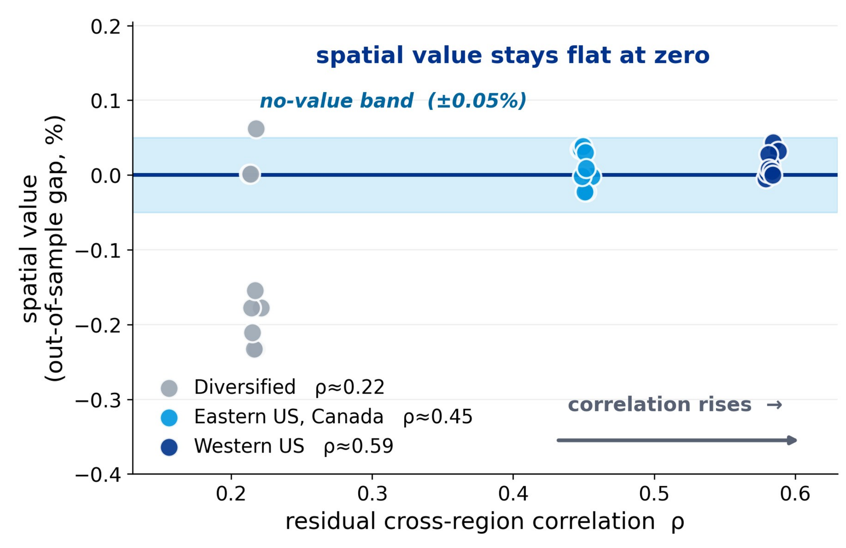
STEP 01 THE PREMISE

The spatial signal is real, but does it pay?

Across 2021 to 2025, de-seasonalized carbon intensity in neighbouring zones moves together. Residual cross-region correlation reaches **0.78** in the Western US and 0.73 in the Eastern interconnection. If spatial structure can ever sharpen a robust schedule, this is the regime where it should.



*Three grids span the US and Canada; the **Western US zones form a tightly correlated graph** (up to 0.78).*



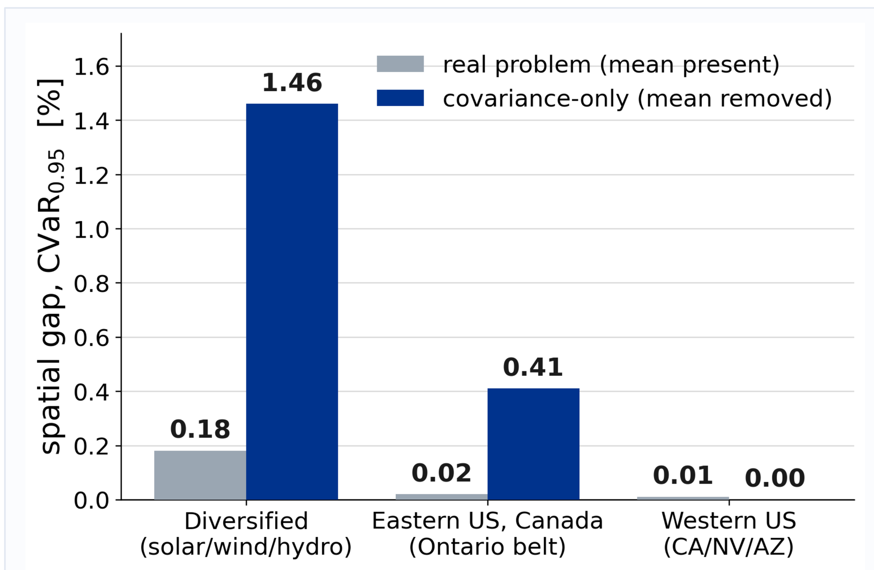
*But value does not follow: **out-of-sample spatial value stays at zero** as correlation rises from 0.2 to 0.6.*

WE EXPECTED	WE FOUND
Correlation sharpens the schedule	**Spatial value stays ~0%**
Copulas capture the dependence	**Even the maximal copula adds nothing**
The signal is simply absent	**Real, but masked: +1.46%**

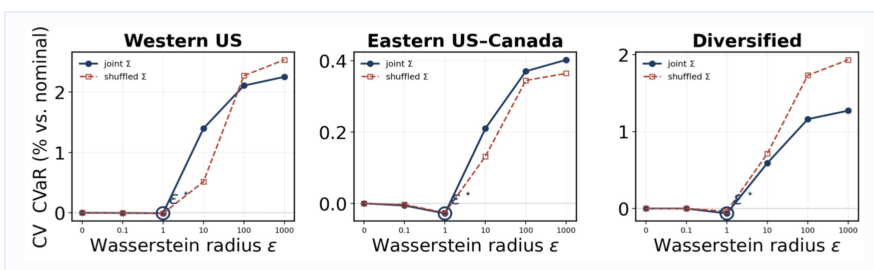
STEP 02 THE SCREEN

Does passive sophistication pay? No.

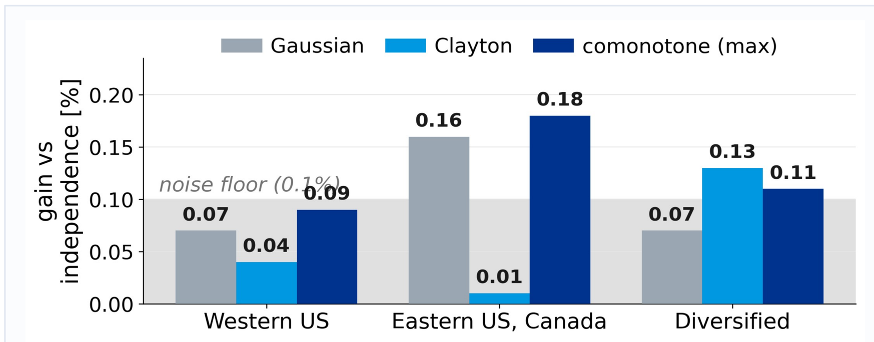
We re-fit the scheduler on the full joint covariance and on a block-diagonal version that erases every cross-region term, then compare out-of-sample CVaR of carbon cost. Across grids and horizons the gap **stays at zero**.



Real problem versus covariance-only: about zero unless the mean field is removed.



Cross-validation picks $\varepsilon^* = 1$: the optimizer truly robustifies, so the null is real.



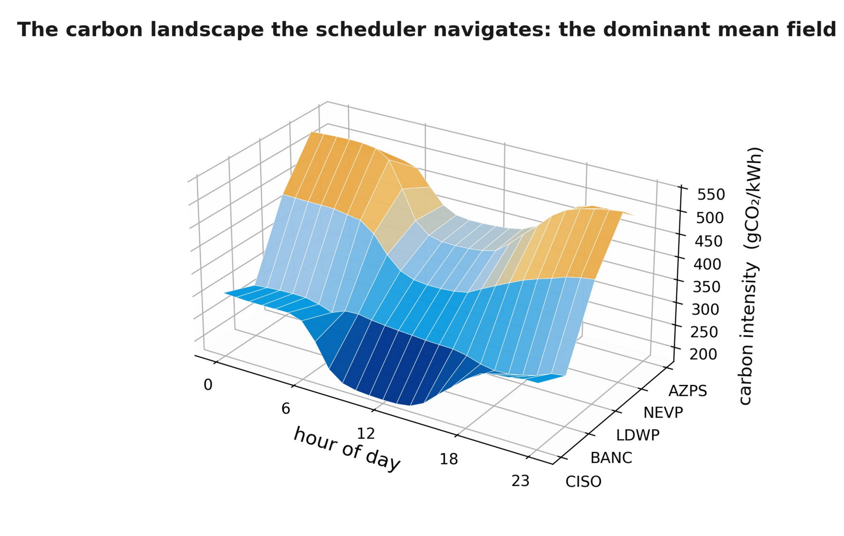
*Gaussian, Clayton, even the comonotone (maximal) copula: **every gain sits at the noise floor.***

THE VERDICT across every estimator and test, the out-of-sample spatial value is statistically indistinguishable from zero.

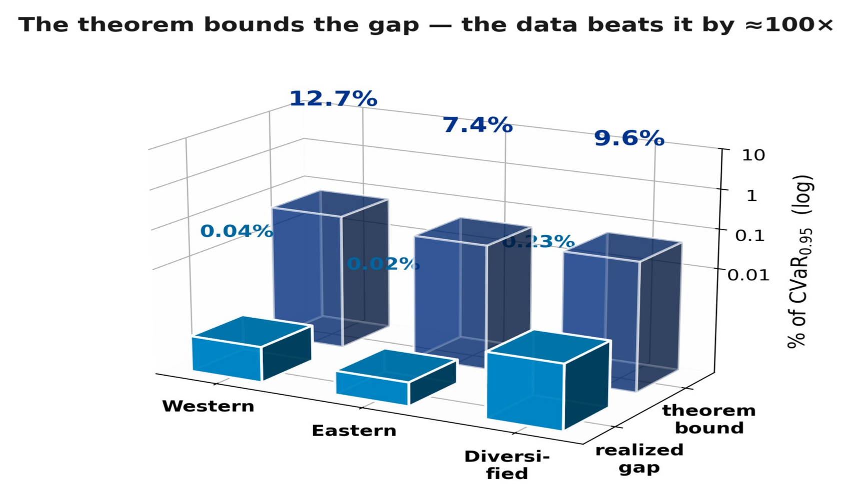
STEP 03 THE MECHANISM

Why does it add nothing? The mean field holds it.

The mean-leveling test isolates the cause. Hold the covariance fixed and flatten the mean field, and the same covariance structure now pays **+1.46%**. The signal is real but masked by a dominant mean, not absent.



*The **dominant mean carbon field**: the schedule follows this terrain; the spatial covariance is a thin ripple on top (≤1.46%).*



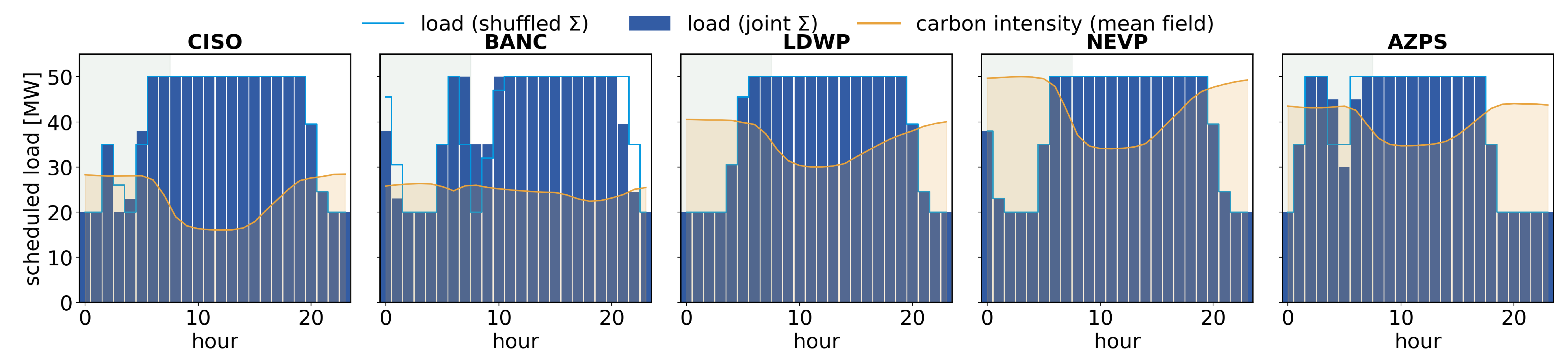
*The mean-dominance theorem caps the gap a-priori at **7 to 13% of CVaR**; the realized gap never exceeds 0.23%, about 100x below its own bound.*

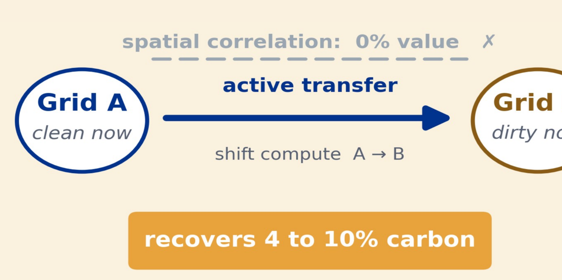
THE NULL SURVIVES EVERY CHECK

- ✓ Ledoit-Wolf covariance shrinkage
- ✓ Residualization of the mean field
- ✓ Benjamini-Hochberg over 144 cells
- ✓ Walk-forward validation to 2024
- ✓ Ramp and utilization sweeps
- ✓ The full CVaR tail range, 0.80 to 0.99

THE RESULT, SEEN DIRECTLY

Each panel is one Western US grid. Bars: scheduled load. Line: carbon intensity. Joint and shuffled plans overlap.





THE PAYOFF

The value is in physically moving compute toward whichever grid is clean right now. An **active inter-region transfer channel** cuts 4 to 10 percent of carbon, deterministically. The sophisticated layers (joint covariance, copulas, distributional robustness) add nothing until grid emergencies hit 3x severity, which real grids never reach. Deploy the simple migration scheduler; price robustness only if your grid faces genuine extremes.

RECOMMENDATION **Deploy the deterministic migration scheduler; skip joint-covariance and copula modeling; price robustness only past a severity real grids never reach.**

REPOSITORY DATA · THESIS

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References: Hall et al. 2024 (carbon-aware DRO); Mohajerin Esfahani and Kuhn 2018 (Wasserstein DRO); Rockafellar and Uryasev 2000 (CVaR).